

ARNAV GUPTA

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EDUCATION

Georgia Institute of Technology

May 2028

Bachelor of Science in Computer Engineering

GPA:3.93

- **Relevant Coursework:** OS, Embedded Systems, Computer Architecture, Data Structures & Algorithms, Processor Design
- **Honors & Awards:** 1st Place / Winner - LinkedIn Intern Hackathon (Summer 2026)

SKILLS

Languages: Python, C, C++, CUDA, Java, SQL, Golang, HTML/CSS, JavaScript, Ruby, TypeScript

Frameworks & Tools: TensorFlow, PyTorch, Linux, FastAPI, Flask, React, Langgraph, LangChain, HuggingFace, Pandas, Node.js, FastMCP, OpenCV, REST APIs, AWS, GCP, .NET, Networking

Developer Tools: Make/CMake, Git, VS Code, PyCharm, Eclipse, Android Studio, Arduino, SolidWorks

Hardware & Low-Level: Parallel Programming (CUDA/OpenMP), GPU/CPU Architecture, VHDL, Raspberry Pi, ESP8266, FPGA design

EXPERIENCE

LinkedIn | Software Engineering Intern

May – August 2026

- Architected an **AI oncall-triage agent** across 4 ML feature platforms, pre-triaging **70% of alerts** and saving ~ **46 eng-hours/year** in MTTR by building a **cheap-before-expensive, parallelized pipeline** that short-circuits simple signals to reduce **triage latency by 45%**.
- Engineered a citable knowledge base with a **self-improving feedback loop**, capturing resolved incident lessons via human-gated PRs to continuously accelerate future triage while **eliminating vector DB overhead** using lazy-loaded context layers and direct git-line citations.

Advanced Networked Systems Research (ANSR) Lab | Undergraduate Researcher

September 2026 – Present

- Extended **LDB (NSDI '24')**, a low-overhead latency profiling system for multithreaded applications, improving **tail-latency root-cause analysis** while reducing **profiling overhead by 22%**.
- Developed systems-level profiling mechanisms using **software-only stack sampling and event tracing** to capture per-function execution latency and **cross-thread interactions** in large-scale applications.

TrueFoundry | Software Engineering Intern

May – August 2025

- On Enterprise Deep Research team with **NVIDIA**, engineered **MCP server** infrastructure and **deepagent-driven workflows** to analyze enterprise datasets, enabling multi-provider LLM deployment and delivering sales teams actionable insights while **reducing query latency by 20%**.
- Built a **Deep Research Pipeline** using **LangGraph** and coordinated **hierarchical sub-agents** to extract and synthesize insights from enterprise data, **achieving 95% accuracy** in meeting intelligence for sales strategy.
- Optimized **TrueFoundry's LLM Gateway** for multi-model routing, enabling low-latency deployments for research workflows.

Mercor | Technical Advisor Intern

January – March 2025

- Developed RLHF pipelines by coding and testing prompt-based solutions, contributing to AI safety and usability research.

PROJECTS

SLM Agent Distillation | Machine Learning Engineer

- Built **LLM-to-SLM distillation pipeline** by logging **50K+ agent traces**, curating datasets, and fine-tuning **2.7B–7B** parameter models with QLoRA, replicating LLM outputs with **92% accuracy**.
- Reduced inference **latency by 8–12×** and compute costs by **85%** through **PEFT fine-tuning, clustering subtasks**, and routing **60%+ agent calls** to SLMs, boosting throughput by **3.5×**.

Coastlens | Full Stack

- Developed a multimodal AI app (Flask) to assess property risk by integrating 12+ years of NOAA hurricane data and Google Earth satellite telemetry.
- Web-scraped **2,000+ articles** from Firecrawl API to provide real-time insights, improving accuracy of recommendations by **25%**.
- Optimized display of relevant news articles with HuggingFace embeddings + Pinecone RAG database, reducing latency by **40%**.

MemoirAI | Full Stack Developer

- Designed scalable cloud architecture on **GCP + MongoDB** for media upload/storage with EXIF metadata and embeddings.
- Integrated **OpenAI CLIP multimodal LLM**, boosting media search **accuracy by 30%** and cutting **retrieval time by 40%**.

Assistive Technology Project: Medication Pillbox (SNAP) | Patent Pending

- Designed and prototyped a pillbox with a C++ controlled Wi-Fi microcontroller to measure and improve medication adherence.
- Achieved a **20% increase** in adherence among elderly patients during pilot testing.
- Partnered with **CVS Health** VP and executed cloud sync via Firestore, scaling to **100K+ potential users** across healthcare networks.